

Deep Dive Tutorial 1: IRRBB Regulatory Framework & Key Metrics

1. Basel III/IV IRRBB Standards

The **Basel Committee on Banking Supervision**

(**BCBS**) introduced **IRRBB** guidelines under **Basel III (BCBS 368, 2016)** and later refinements in **Basel IV (2023)**. These standards require banks to measure and manage interest rate risk in the banking book (IRRBB) through two primary approaches:

- **Earnings-Based Approach (NII at Risk):** Measures the impact of interest rate changes on **Net Interest Income (NII)** over a short-term horizon (typically 1-2 years). Banks must assess NII sensitivity under regulatory shock scenarios (e.g., +200bps, -200bps).
- **Economic Value-Based Approach (EVE at Risk):** Evaluates the long-term impact on a bank's **Economic Value of Equity (EVE)** by discounting future cash flows under stressed yield curves. EVE measures the net present value (NPV) of banking book assets minus liabilities.

Key Regulatory References:

- **BCBS 368 (2016)** – Standards for IRRBB
- **EBA/GL/2018/02** – IRRBB supervisory guidelines
- **FRB SR 10-1 / SR 15-18** – U.S. regulatory expectations

2. Earnings-Based Approach (NII at Risk)

Net Interest Income (NII) Sensitivity measures how changes in interest rates affect a bank's interest earnings. Banks must model:

- **Parallel Shocks:** Uniform shifts in the yield curve (e.g., +100bps across all maturities).

- **Non-Parallel Shocks:** Twists or steepening/flattening of the yield curve (e.g., short-term rates rise while long-term rates fall).

Static vs. Dynamic Balance Sheet Assumptions:

- **Static Approach:** Assumes no new business or balance sheet growth (simpler but less realistic).
- **Dynamic Approach:** Incorporates new loans, deposits, and hedging strategies (more complex but accurate).

Hands-on Exercise:

- Simulate NII impact under **+100bps and -100bps shocks** using a simplified banking book.
- Compare static vs. dynamic balance sheet assumptions.

3. Economic Value-Based Approach (EVE at Risk)

Economic Value of Equity (EVE) represents the NPV of a bank's assets minus liabilities. The **DCF approach** is used:

$$EVE = \sum (CF_{Assets} / (1+r)^t) - \sum (CF_{Liabilities} / (1+r)^t) \quad EVE = \sum ((1+r)^{-t} CF_{Assets}) - \sum ((1+r)^{-t} CF_{Liabilities})$$

Key Considerations:

- **Discount Rate Selection:** Banks use risk-free rates (e.g., OIS) plus commercial spreads.
- **Behavioural Adjustments:**
 - **Prepayment Risk:** Mortgage borrowers refinance if rates fall.
 - **Deposit Stickiness:** Non-maturity deposits (NMDs) may not reprice immediately.
 - **Optionality:** Embedded caps/floors in loans/deposits alter cash flows.

Hands-on Exercise:

- Compute EVE for a simplified banking book (fixed-rate loans + floating-rate deposits).
- Assess EVE sensitivity under **steepening/flattening yield curve scenarios**.

4. Impact of Optionality (Prepayments, Caps/Floors)

Optionality introduces **non-linear risks** in IRRBB:

- **Mortgage Prepayments:** When rates fall, borrowers refinance, shortening asset duration.
- **Deposit Withdrawal Risk:** If rates rise, depositors may shift to higher-yielding accounts.
- **Caps/Floors in Loans:** Floating-rate loans with rate ceilings (caps) limit upside in rising rate environments.

Modeling Techniques:

- **Option-Adjusted Spread (OAS) Models** for embedded optionality.
- **Monte Carlo Simulations** for stochastic prepayment behavior.

Regulatory Example:

- **BCBS requires** banks to model **behavioral assumptions** (e.g., 20% deposit beta under shocks).

Conclusion & Next Steps

- **Regulatory IRRBB modeling** requires **NII & EVE sensitivity analysis** under prescribed shocks.
- **Dynamic balance sheet modeling** is increasingly preferred for accuracy.
- **Optionality risks** must be quantified using advanced techniques (OAS, Monte Carlo).

Further Reading:

- **BCBS 368 (2016):** [Basel IRRBB Standards](#)
- **EBA Guidelines:** [EBA/GL/2018/02](#)
- **FRB SR 15-18:** [U.S. IRRBB Guidance](#)